Docker Back End Deployment Process (Part-1)

Step-1: Download project which was sent to your email and save it on Desktop

Step-2: Open the same in VS code

CMD -> code .

Step-3: Build and Run Docker image (Using dockerfile.one)

```
FROM maven:3.9-eclipse-temurin-21 AS build
WORKDIR /app
COPY pom.xml .
RUN --mount=type=cache,target=/root/.m2 mvn -B -DskipTests dependency:go-offline
COPY src ./src
RUN --mount=type=cache,target=/root/.m2 mvn -B -DskipTests clean install package

FROM eclipse-temurin:21-jre-alpine
WORKDIR /app
COPY --from=build /app/target/*.jar app.jar
ENV JAVA_OPTS=""
EXPOSE 8081
ENTRYPOINT ["sh","-c","java $JAVA_OPTS -jar app.jar"]
```

NOTE: You can place this code also in a file named as **dockerfile**

Step-4: Define Docker YML File (docker-compose.yml)

a. If your docker file is named as "dockerfile.one" in step-3 then place the following code in docker-compose.yml

```
services:
 db:
  image: mysql:8.0
  container_name: ecommerce-db
  environment:
   MYSQL_ROOT_PASSWORD: root
   MYSQL_DATABASE: ecommerce
  healthcheck:
   test: ["CMD", "mysqladmin", "ping", "-h", "127.0.0.1", "-uroot", "-proot"]
   interval: 5s
   timeout: 3s
   retries: 20

 backend:
  build:
    context: .
    dockerfile: dockerfile.one
  container_name: ecommerce-backend
  depends_on:
   db:
    condition: service_healthy
  environment:
   SERVER_PORT: 8081
   SPRING_DATASOURCE_URL:
   jdbc:mysql://db:3306/ecommerce?useSSL=false&allowPublicKeyRetrieval=tru
   e&serverTimezone=UTC
   SPRING_DATASOURCE_USERNAME: root
   SPRING_DATASOURCE_PASSWORD: root
  ports:
   - "8081:8081"
     restart: unless-stopped
```

(OR)

b. If your docker file is named as “dockerfile” in step-3 then place the following code in docker-compose.yml

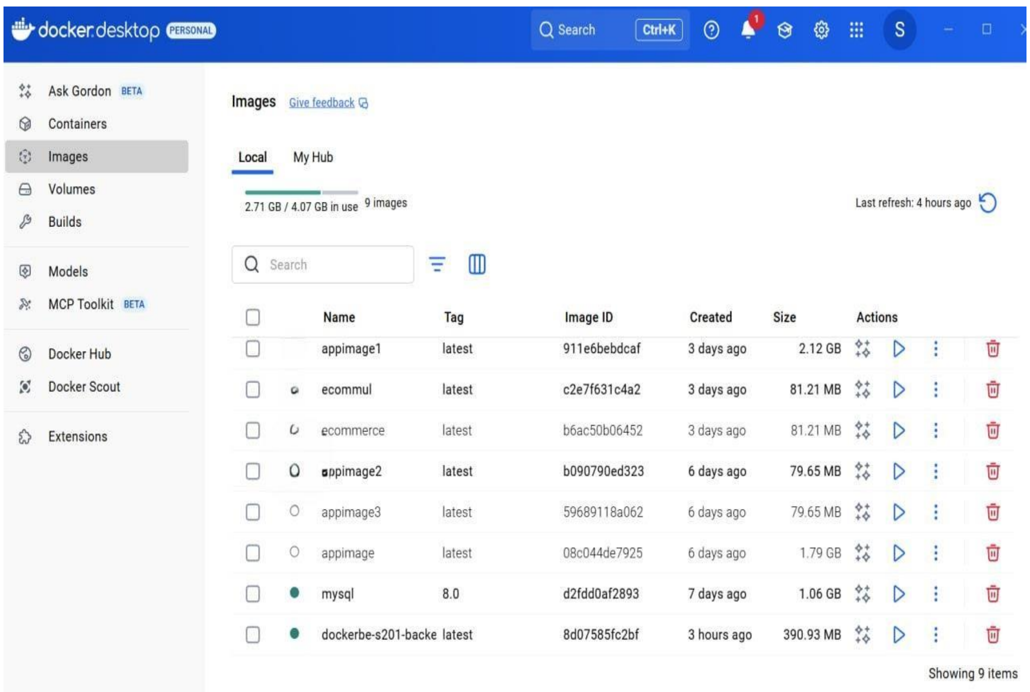
```yaml
services:
 db:
  image: mysql:8.0
  container_name: ecommerce-db
  environment:
   MYSQL_ROOT_PASSWORD: root
   MYSQL_DATABASE: ecommerce
  healthcheck:
   test: ["CMD", "mysqladmin", "ping", "-h", "127.0.0.1", "-uroot", "-proot"]
   interval: 5s
   timeout: 3s
   retries: 20

 backend:
  build: .
  container_name: ecommerce-backend
  depends_on:
   db:
    condition: service_healthy
  environment:
   SERVER_PORT: 8081
   SPRING_DATASOURCE_URL:
   jdbc:mysql://db:3306/ecommerce?useSSL=false&allowPublicKeyRetrieval=tru
   e&serverTimezone=UTC
   SPRING_DATASOURCE_USERNAME: root
   SPRING_DATASOURCE_PASSWORD: root
  ports:
   - "8081:8081"
     restart: unless-stopped
```

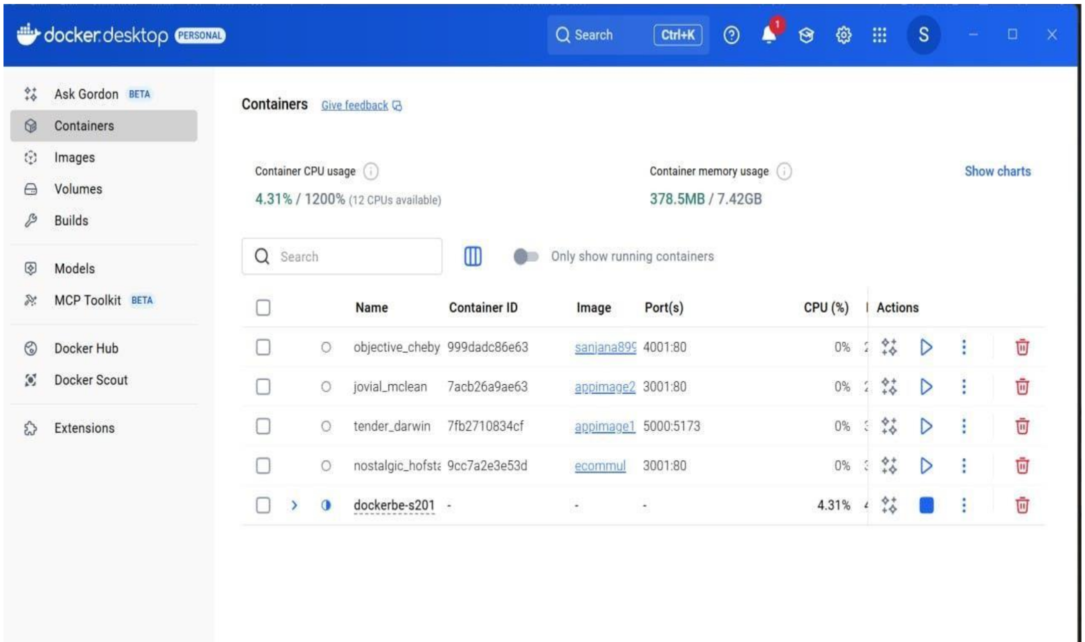
Step-5: Build Docker Image

- Open VS Code terminal and make sure you are in command prompt
- Now run the following command to build the image. Once it is successfully executed, two images will be created as shown below
- COMMAND:

  docker compose up -d –build



- Now check your Docker Container, the image in container will be created as shown below



When you click on dockerbe-s104